

ORCA

Scientific Ocean & Fishing Reasoning Rulebook

SIH26176 — Marine EcOsystem Reasoning with Collaborative Agents

Purpose

Provide the coding/AI agent with deterministic scientific rules for ocean analysis and fishing-zone reasoning, while reducing LLM hallucinations and improving the accuracy of information displayed visually on the map.

CORE PRINCIPLE

The scientific backend calculates measurements, features, scores, coordinates, routes and geometries. The LLM explains those verified results in simple language. The LLM must never invent scientific values or map geometry.

1. Core Architecture

MOSDAC / Satellite + Copernicus Physics + Copernicus Biogeochemistry + INCOIS PFZ + IMD → Scientific Processing Engine → Physical Ocean Features → Fishing Suitability Model → Safety Filter → Route Engine → Map → LLM → Simple Human Answer.

LLM receives compact, validated scientific results rather than raw large scientific files.

2. SST Gradient

$$|\nabla \text{SST}| = \text{sqrt}((\partial \text{SST} / \partial x)^2 + (\partial \text{SST} / \partial y)^2)$$

Interpretation: how rapidly sea temperature changes across an area. Use it as evidence for thermal fronts.

Never calculate gradients across land. Use proper geographic distance for x/y.

3. Thermal Front Detection

Use Cayula–Cornillon-style thermal-front detection where appropriate, preferably with regionally validated INCOIS parameters.

Do not invent thresholds. Record algorithm, parameters, dataset, resolution and timestamp for every calculation.

Do not blindly reuse a threshold from one sensor or resolution for another dataset.

4. Chlorophyll

Use chlorophyll-a as biological productivity evidence.

Where appropriate for skewed distributions, use $\log_{10}(\text{CHL} + \epsilon)$ before normalization.

Normalize relative to local region and season when building suitability features.

High chlorophyll does not mean guaranteed fish.

5. Chlorophyll Fronts

Use validated edge detection, including Canny-based detection where consistent with the INCOIS methodology.

Combine meaningful SST and chlorophyll fronts as supporting PFZ evidence.

6. Front Persistence

Persistence = feature-present valid observations / valid observations examined.

Use multiple valid observations. A one-time feature should not automatically be treated as a persistent PFZ.

Persistence is especially useful when combined with SST gradient and chlorophyll.

7. Ocean Currents — Copernicus

Represent surface current as $V=(u,v)$, where u is eastward velocity and v is northward velocity.

$$\text{Speed} = \text{sqrt}(u^2 + v^2)$$

$$\text{Direction} = \text{atan2}(v,u)$$

Follow the Copernicus dataset metadata for sign and direction conventions.

8. Relative Vorticity

$$\zeta = \partial v / \partial x - \partial u / \partial y$$

Use it as an ocean-dynamics / eddy indicator. Do not interpret vorticity directly as fish probability.

9. Okubo–Weiss Eddy Indicator

$$S_n = \partial u / \partial x - \partial v / \partial y$$

$$S_s = \partial v / \partial x + \partial u / \partial y$$

$$W = S_n^2 + S_s^2 - \zeta^2$$

$W < 0$ indicates rotation-dominated flow; $W > 0$ indicates strain-dominated flow. Treat this as feature evidence, not direct fish probability.

10. Sea-Surface Height

Where appropriate: SSH anomaly = SSH – SSH climatology.

Use SSH structure together with currents, SST and other features to identify mesoscale ocean structures.

Prefer a Copernicus-provided derived geostrophic-current variable when appropriate instead of recalculating it unnecessarily.

11. Geostrophic Current

$$f = 2\Omega \sin(\text{latitude})$$

$$u_g = -(g/f)(\partial \eta / \partial y)$$

$$v_g = (g/f)(\partial \eta / \partial x)$$

Do not use the simple geostrophic approximation near the equator without special handling. Use dataset metadata and native derived variables when available.

12. Upwelling

Use multiple indicators: SST anomaly + wind + Ekman transport + vertical velocity + chlorophyll + circulation.

$$\text{Wind stress: } \tau = \text{pair CD } |U| U$$

Ekman transport is approximately related to $\tau / (\rho_w f)$, with direction depending on hemisphere and coastline orientation.

Do not classify upwelling from SST alone. Follow dataset-specific sign conventions.

13. Vertical Velocity

Where Copernicus provides vertical velocity, use the product variable rather than deriving it unnecessarily from noisy fields.

Follow the dataset's documented sign convention for upward/downward velocity.

14. Mixed Layer Depth

Use Copernicus MLD as evidence about vertical mixing and stratification.

Do not treat MLD directly as fish abundance. Follow the product's documented MLD definition.

15. Copernicus Biogeochemistry

Where available, use chlorophyll, phytoplankton, primary production, nitrate, phosphate, oxygen and related variables as supporting ecological evidence.

Respect native spatial resolution. Do not display coarse BGC model output as high-resolution local information.

16. Fishing Model

Do not invent arbitrary weights and call them scientifically accurate.

First use validated INCOIS PFZ methodology.

Where sufficient geo-tagged catch/landing ground truth exists, calibrate a supervised model using environmental predictors and fishing observations.

Possible outputs: low / medium / high suitability, or a calibrated probability of favourable fishing conditions.

Do not claim exact fish quantity unless ground-truth catch data supports such a prediction.

17. Safety Model

Fishing suitability and safety are separate decisions.

Safety must override fishing suitability. A highly productive location with dangerous conditions must not be recommended.

Use weather, waves, wind, lightning, cyclone information, warnings, route exposure and other relevant safety constraints.

18. Data Quality

Every variable should retain: source, dataset, unit, timestamp, forecast time, retrieval time, resolution, quality flag, missing flag and uncertainty where available.

Never replace missing values with zero.

Apply land, cloud and quality masks before scientific calculations.

19. Spatial Accuracy

The LLM must never create coordinates. Scientific services produce geometry; the map displays backend-generated geometry.

Maintain native data resolution. Record every resampling operation.

Never interpolate across land.

20. Temporal Accuracy

Distinguish observation, analysis, forecast, reanalysis and historical data.

Never describe historical or reanalysis data as live.

For 'now/current/safe now' queries, perform a fresh data-availability check.

21. Source Agreement

Compare independent sources where possible.

Agreement can increase confidence; disagreement should reduce confidence or trigger further validation.

Never hide major source disagreement.

22. LLM Anti-Hallucination Rules

LLM must NOT calculate or invent scientific measurements.

LLM must NOT invent coordinates, routes, confidence scores or timestamps.

LLM must NOT claim data is live when it is not.

LLM only explains deterministic scientific results in simple language.

23. User Output

Do not expose unnecessary scientific readings.

Prefer conclusions such as: 'Best fishing area found.' 'Conditions are favourable.' 'Sea conditions are unsafe.' 'This route has lower assessed risk.' 'Live data is unavailable.'

Technical scientific details should appear only in an optional Details / Sources section.

24. Recommended ORCA Decision Pipeline

SST + CHL + INCOIS PFZ → SST gradient / thermal fronts → chlorophyll fronts → front persistence → currents / SSH / MLD → eddy / upwelling indicators → fishing suitability → IMD safety filter → route safety → final recommendation → backend geometry → map → simple LLM explanation.

Keep the scientific engine deterministic and auditable. Keep the LLM as the explanation and conversation layer.